

Discrimination and Male – Female Earnings Differentials

ECON 300 - Labour Economics

Muhebullah Karimzada

Winter 2026

- Describe the factors giving rise to male–female pay gaps and the role of discrimination.
- Outline methodologies used to estimate discriminatory pay gaps and their challenges.
- Explain how discriminatory gaps can persist even with competitive pressures.
- Discuss pros and cons of policy initiatives to combat discrimination.
- Explain the evolution of the gender pay gap and the mechanics of pay equity.

- Describe the factors giving rise to male–female pay gaps and the role of discrimination.
- Outline methodologies used to estimate discriminatory pay gaps and their challenges.
- Explain how discriminatory gaps can persist even with competitive pressures.
- Discuss pros and cons of policy initiatives to combat discrimination.
- Explain the evolution of the gender pay gap and the mechanics of pay equity.

Learning Objectives

- Describe the factors giving rise to male–female pay gaps and the role of discrimination.
- Outline methodologies used to estimate discriminatory pay gaps and their challenges.
- Explain how discriminatory gaps can persist even with competitive pressures.
- Discuss pros and cons of policy initiatives to combat discrimination.
- Explain the evolution of the gender pay gap and the mechanics of pay equity.

Learning Objectives

- Describe the factors giving rise to male–female pay gaps and the role of discrimination.
- Outline methodologies used to estimate discriminatory pay gaps and their challenges.
- Explain how discriminatory gaps can persist even with competitive pressures.
- Discuss pros and cons of policy initiatives to combat discrimination.
- Explain the evolution of the gender pay gap and the mechanics of pay equity.

- Describe the factors giving rise to male–female pay gaps and the role of discrimination.
- Outline methodologies used to estimate discriminatory pay gaps and their challenges.
- Explain how discriminatory gaps can persist even with competitive pressures.
- Discuss pros and cons of policy initiatives to combat discrimination.
- Explain the evolution of the gender pay gap and the mechanics of pay equity.

- How can equally productive men and women be paid different wages in competitive markets?
- Will competitive forces dissipate discrimination?
- How do we measure discrimination credibly in the data?
- How much discrimination exists in Canada? What about other groups?
- Which policies have been adopted, and which are effective?

- How can equally productive men and women be paid different wages in competitive markets?
- Will competitive forces dissipate discrimination?
- How do we measure discrimination credibly in the data?
- How much discrimination exists in Canada? What about other groups?
- Which policies have been adopted, and which are effective?

- How can equally productive men and women be paid different wages in competitive markets?
- Will competitive forces dissipate discrimination?
- How do we measure discrimination credibly in the data?
- How much discrimination exists in Canada? What about other groups?
- Which policies have been adopted, and which are effective?

- How can equally productive men and women be paid different wages in competitive markets?
- Will competitive forces dissipate discrimination?
- How do we measure discrimination credibly in the data?
- How much discrimination exists in Canada? What about other groups?
- Which policies have been adopted, and which are effective?

- How can equally productive men and women be paid different wages in competitive markets?
- Will competitive forces dissipate discrimination?
- How do we measure discrimination credibly in the data?
- How much discrimination exists in Canada? What about other groups?
- Which policies have been adopted, and which are effective?

What is Labour-Market Discrimination?

- Unequal treatment of groups with the same productivity (or productivity-related traits).
- Who can discriminate?
 - Employers: wages, hiring, promotions
 - Co-workers: prejudice, misinformation, job-security concerns
 - Customers: discriminatory preferences

What is Labour-Market Discrimination?

- Unequal treatment of groups with the same productivity (or productivity-related traits).
- **Who can discriminate?**
 - **Employers:** wages, hiring, promotions
 - **Co-workers:** prejudice, misinformation, job-security concerns
 - **Customers:** discriminatory preferences

What is Labour-Market Discrimination?

- Unequal treatment of groups with the same productivity (or productivity-related traits).
- **Who can discriminate?**
 - **Employers:** wages, hiring, promotions
 - **Co-workers:** prejudice, misinformation, job-security concerns
 - **Customers:** discriminatory preferences

What is Labour-Market Discrimination?

- Unequal treatment of groups with the same productivity (or productivity-related traits).
- **Who can discriminate?**
 - **Employers:** wages, hiring, promotions
 - **Co-workers:** prejudice, misinformation, job-security concerns
 - **Customers:** discriminatory preferences

What is Labour-Market Discrimination?

- Unequal treatment of groups with the same productivity (or productivity-related traits).
- **Who can discriminate?**
 - **Employers:** wages, hiring, promotions
 - **Co-workers:** prejudice, misinformation, job-security concerns
 - **Customers:** discriminatory preferences

- **Demand theories**
- Supply theories
- Noncompetitive theories
- Queuing and efficiency-wage considerations

- Demand theories
- Supply theories
- Noncompetitive theories
- Queuing and efficiency-wage considerations

- Demand theories
- Supply theories
- Noncompetitive theories
- Queuing and efficiency-wage considerations

- Demand theories
- Supply theories
- Noncompetitive theories
- Queuing and efficiency-wage considerations

- Lower **demand** for female labour $\Rightarrow$ lower wages and employment.
- Possible channels:
 - Underestimation of female productivity
 - Statistical discrimination / signalling stories

- Lower **demand** for female labour $\Rightarrow$ lower wages and employment.
- Possible channels:
 - Underestimation of female productivity
 - Statistical discrimination / signalling stories

- Lower **demand** for female labour $\Rightarrow$ lower wages and employment.
- Possible channels:
 - Underestimation of female productivity
 - Statistical discrimination / signalling stories

- Lower **demand** for female labour $\Rightarrow$ lower wages and employment.
- Possible channels:
 - Underestimation of female productivity
 - Statistical discrimination / signalling stories

- Increased **supply** of female labour depresses wages if productivity or matching suffers.
- **Crowding hypothesis**: Segregation into “female-type” jobs $\Rightarrow$ excess supply $\Rightarrow$ lower MRP and wages.
- **Dual labour market**: Men in core, women in secondary sector $\Rightarrow$ lower wages in secondary market.

Supply Theories of Discrimination

- Increased **supply** of female labour depresses wages if productivity or matching suffers.
- **Crowding hypothesis:** Segregation into “female-type” jobs $\Rightarrow$ excess supply $\Rightarrow$ lower MRP and wages.
- **Dual labour market:** Men in core, women in secondary sector $\Rightarrow$ lower wages in secondary market.

- Increased **supply** of female labour depresses wages if productivity or matching suffers.
- **Crowding hypothesis:** Segregation into “female-type” jobs $\Rightarrow$ excess supply $\Rightarrow$ lower MRP and wages.
- **Dual labour market:** Men in core, women in secondary sector $\Rightarrow$ lower wages in secondary market.

- **Noncompetitive:** Adjustment costs and imperfect information can sustain gaps.
- **Queuing/Efficiency wages:** Firms pay above-competitive wages to reduce turnover and improve morale, creating queues that can interact with group differences.

- **Noncompetitive:** Adjustment costs and imperfect information can sustain gaps.
- **Queuing/Efficiency wages:** Firms pay above-competitive wages to reduce turnover and improve morale, creating queues that can interact with group differences.

Productivity Differences: Choice or Discrimination?

- Differences in acquired human capital may reflect:
 - Shorter expected labour-force attachment
 - Higher turnover or absenteeism
 - Household and caregiving responsibilities

Productivity Differences: Choice or Discrimination?

- Differences in acquired human capital may reflect:
 - Shorter expected labour-force attachment
 - Higher turnover or absenteeism
 - Household and caregiving responsibilities

Productivity Differences: Choice or Discrimination?

- Differences in acquired human capital may reflect:
 - Shorter expected labour-force attachment
 - Higher turnover or absenteeism
 - Household and caregiving responsibilities

Productivity Differences: Choice or Discrimination?

- Differences in acquired human capital may reflect:
 - Shorter expected labour-force attachment
 - Higher turnover or absenteeism
 - Household and caregiving responsibilities

Setup: Oaxaca(-Blinder) Decomposition

Let $\ln W = X\beta + \varepsilon$, estimated separately for men (m) and women (f).

$$\overline{\ln W}_m - \overline{\ln W}_f = (\overline{X}_m - \overline{X}_f)^\top \hat{\beta}_m + \overline{X}_f^\top (\hat{\beta}_m - \hat{\beta}_f)$$

- First term: part “explained” by differences in characteristics ($\overline{X}$).
- Second term: **coefficient effect** (“unexplained”/often attributed to discrimination).
- Reference structure can be male, female, or pooled (affects magnitudes).

Setup: Oaxaca(-Blinder) Decomposition

Let $\ln W = X\beta + \varepsilon$, estimated separately for men (m) and women (f).

$$\overline{\ln W}_m - \overline{\ln W}_f = (\overline{X}_m - \overline{X}_f)^\top \hat{\beta}_m + \overline{X}_f^\top (\hat{\beta}_m - \hat{\beta}_f)$$

- First term: part “explained” by differences in characteristics ($\overline{X}$).
- Second term: **coefficient effect** (“unexplained”/often attributed to discrimination).
- Reference structure can be male, female, or pooled (affects magnitudes).

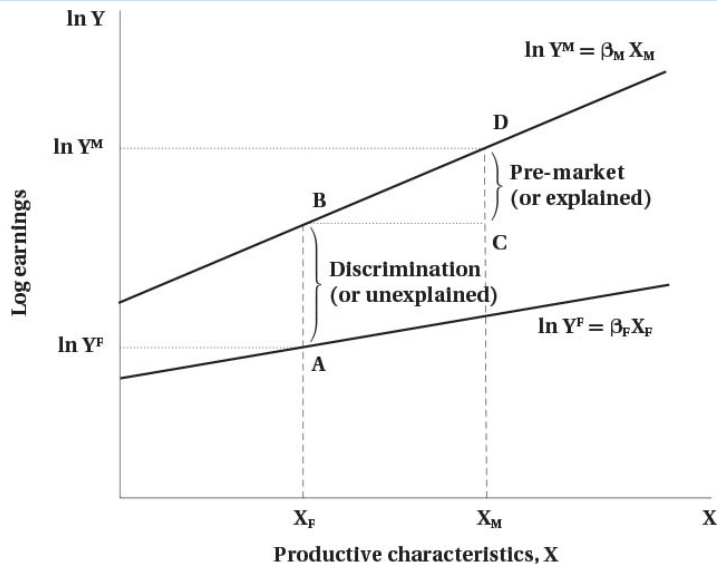
Setup: Oaxaca(-Blinder) Decomposition

Let $\ln W = X\beta + \varepsilon$, estimated separately for men (m) and women (f).

$$\overline{\ln W}_m - \overline{\ln W}_f = (\overline{X}_m - \overline{X}_f)^\top \hat{\beta}_m + \overline{X}_f^\top (\hat{\beta}_m - \hat{\beta}_f)$$

- First term: part “explained” by differences in characteristics ($\overline{X}$).
- Second term: **coefficient effect** (“unexplained”/often attributed to discrimination).
- Reference structure can be male, female, or pooled (affects magnitudes).

Figure 12.1: Graphical Illustration of the Oaxaca Decomposition



Worked Example: Analyzing the Determinants of Earnings

Estimated earnings equations:

$$\text{Men: } \ln W_m = 0.40 + 0.07 S + 0.06 \text{ EXP} + 0.05 \text{ Married} + 0.01 \text{ Children}$$

$$\text{Women: } \ln W_f = 0.01 + 0.10 S + 0.04 \text{ EXP} - 0.02 \text{ Married} - 0.05 \text{ Children}$$

Group means:

$$\bar{X}_m = (S=14, \text{EXP}=10, M=0.5, C=0.6), \bar{X}_f = (S=15, \text{EXP}=7, M=0.5, C=0.6).$$

Worked Example: Predicted Values and Raw Gap

$$E(\ln W_m) = 0.40 + 0.07(14) + 0.06(10) + 0.05(0.5) + 0.01(0.6) = \mathbf{2.01}$$

$$E(\ln W_f) = 0.01 + 0.10(15) + 0.04(7) - 0.02(0.5) - 0.05(0.6) = \mathbf{1.75}$$

Raw ratio (female to male):

$$\frac{E(\ln W_f)}{E(\ln W_m)} = \frac{1.75}{2.01} \approx \mathbf{0.87}.$$

Worked Example: Counterfactual Pay Structures

Female characteristics, **male** coefficients:

$$\ln W_f^* = 0.40 + 0.07(15) + 0.06(7) + 0.05(0.5) + 0.01(0.6) = \mathbf{1.90}$$

Male characteristics, **female** coefficients:

$$\ln W_m^* = 0.01 + 0.10(14) + 0.04(10) - 0.02(0.5) - 0.05(0.6) = \mathbf{1.77}$$

Implications:

$$\ln W_f^* - E(\ln W_m) = 1.90 - 2.01 = -\mathbf{0.11}, \quad E(\ln W_f) - \ln W_m^* = 1.75 - 1.77 = -\mathbf{0.02}.$$

Additional Evidence on Male–Female Pay Differentials

- Studies examine wage, sex, institutional, and household discrimination channels.
- Average female earnings often reported at $\sim 60\text{--}65\%$ of male earnings in some datasets/periods, reflecting barriers (e.g., glass ceilings) and composition effects.
- Key question: **What would women earn if they were men?** Use counterfactuals holding productive traits fixed.

Additional Evidence on Male–Female Pay Differentials

- Studies examine wage, sex, institutional, and household discrimination channels.
- Average female earnings often reported at $\sim 60\text{--}65\%$ of male earnings in some datasets/periods, reflecting barriers (e.g., glass ceilings) and composition effects.
- Key question: *What would women earn if they were men?* Use counterfactuals holding productive traits fixed.

Additional Evidence on Male–Female Pay Differentials

- Studies examine wage, sex, institutional, and household discrimination channels.
- Average female earnings often reported at $\sim 60\text{--}65\%$ of male earnings in some datasets/periods, reflecting barriers (e.g., glass ceilings) and composition effects.
- Key question: **What would women earn if they were men?** Use counterfactuals holding productive traits fixed.

Empirical Evidence for Other Groups

Table 12.1: Ethnic Wage Differentials, Various Groups Born in Canada or United States

	Black	South Asian^a	Southeast Asian^b	Chinese	Aboriginal
<i>Canada</i>					
Total gap	0.293	0.536	−0.022	0.193	0.179
“Discrimination”	0.171	0.222	0.023	0.058	0.091
<i>United States</i>					
Total gap	0.328	0.176	−0.079	−0.205	0.304
“Discrimination”	0.123	0.055	−0.007	−0.047	0.136

Notes: Column headers match the original table; ^aSouth Asian, ^bSoutheast Asian.

- **Conventional equal pay:** Same job, same establishment.
- **Equal value / Pay equity / Comparable worth:** Job evaluation to compare across predominantly male vs. female jobs.
- **Equal employment opportunity (EEO):** Recruitment, hiring, promotion protections.
- **Affirmative action / Employment equity:** Targeted plans for designated groups.
- **Facilitating policies:** Childcare, family-friendly practices to broaden choices.

- **Conventional equal pay:** Same job, same establishment.
- **Equal value / Pay equity / Comparable worth:** Job evaluation to compare across predominantly male vs. female jobs.
- **Equal employment opportunity (EEO):** Recruitment, hiring, promotion protections.
- **Affirmative action / Employment equity:** Targeted plans for designated groups.
- **Facilitating policies:** Childcare, family-friendly practices to broaden choices.

- **Conventional equal pay:** Same job, same establishment.
- **Equal value / Pay equity / Comparable worth:** Job evaluation to compare across predominantly male vs. female jobs.
- **Equal employment opportunity (EEO):** Recruitment, hiring, promotion protections.
- **Affirmative action / Employment equity:** Targeted plans for designated groups.
- **Facilitating policies:** Childcare, family-friendly practices to broaden choices.

- **Conventional equal pay:** Same job, same establishment.
- **Equal value / Pay equity / Comparable worth:** Job evaluation to compare across predominantly male vs. female jobs.
- **Equal employment opportunity (EEO):** Recruitment, hiring, promotion protections.
- **Affirmative action / Employment equity:** Targeted plans for designated groups.
- **Facilitating policies:** Childcare, family-friendly practices to broaden choices.

- **Conventional equal pay:** Same job, same establishment.
- **Equal value / Pay equity / Comparable worth:** Job evaluation to compare across predominantly male vs. female jobs.
- **Equal employment opportunity (EEO):** Recruitment, hiring, promotion protections.
- **Affirmative action / Employment equity:** Targeted plans for designated groups.
- **Facilitating policies:** Childcare, family-friendly practices to broaden choices.

- **Equal pay legislation:** Equal pay for equal *value* within the same establishment; job evaluation must be gender-bias-free.
- Scope: All Canadian jurisdictions except Alberta.

- **Equal pay legislation:** Equal pay for equal *value* within the same establishment; job evaluation must be gender-bias-free.
- Scope: All Canadian jurisdictions except Alberta.

- Define gender predominance and choose job-evaluation procedure.
- Specify adjustment procedures and possible exemptions.
- Scope can be large (cross-occupation comparisons), but enforcement and complaints may be limited in practice.

Comparable Worth: Design Features

- Define gender predominance and choose job-evaluation procedure.
- Specify adjustment procedures and possible exemptions.
- Scope can be large (cross-occupation comparisons), but enforcement and complaints may be limited in practice.

- Define gender predominance and choose job-evaluation procedure.
- Specify adjustment procedures and possible exemptions.
- Scope can be large (cross-occupation comparisons), but enforcement and complaints may be limited in practice.

- EEO: Prevents discrimination in recruiting/hiring/promotion; can raise female employment and wages; complements equal pay.
- Employment equity: Federal jurisdiction; ~ 10% of labour force; designated groups include women, visible minorities, persons with disabilities, and Aboriginal people.

- EEO: Prevents discrimination in recruiting/hiring/promotion; can raise female employment and wages; complements equal pay.
- Employment equity: Federal jurisdiction; ~ 10% of labour force; designated groups include women, visible minorities, persons with disabilities, and Aboriginal people.

- **Equal pay / EEO:** Mixed evidence on broad wage-gap closures.
- **Comparable worth:** Evidence of gap reductions where implemented.
- **Bottom line:** Impacts exist but can be limited by scope, design, and enforcement.

- **Equal pay / EEO:** Mixed evidence on broad wage-gap closures.
- **Comparable worth:** Evidence of gap reductions where implemented.
- **Bottom line:** Impacts exist but can be limited by scope, design, and enforcement.

- **Equal pay / EEO:** Mixed evidence on broad wage-gap closures.
- **Comparable worth:** Evidence of gap reductions where implemented.
- **Bottom line:** Impacts exist but can be limited by scope, design, and enforcement.

- Defined discrimination, reviewed demand/supply/noncompetitive/queuing theories.
- Implemented Oaxaca-type decompositions to separate endowments vs. coefficients.
- Reviewed evidence on gender and other-group differentials, including exact table values.
- Assessed policy tools: equal pay, pay equity, EEO, employment equity, and facilitating policies.

- Defined discrimination, reviewed demand/supply/noncompetitive/queuing theories.
- Implemented Oaxaca-type decompositions to separate endowments vs. coefficients.
- Reviewed evidence on gender and other-group differentials, including exact table values.
- Assessed policy tools: equal pay, pay equity, EEO, employment equity, and facilitating policies.

- Defined discrimination, reviewed demand/supply/noncompetitive/queuing theories.
- Implemented Oaxaca-type decompositions to separate endowments vs. coefficients.
- Reviewed evidence on gender and other-group differentials, including exact table values.
- Assessed policy tools: equal pay, pay equity, EEO, employment equity, and facilitating policies.

- Defined discrimination, reviewed demand/supply/noncompetitive/queuing theories.
- Implemented Oaxaca-type decompositions to separate endowments vs. coefficients.
- Reviewed evidence on gender and other-group differentials, including exact table values.
- Assessed policy tools: equal pay, pay equity, EEO, employment equity, and facilitating policies.

Thank You!